

ABSTRACT

The Construction Workflow Automation System is a comprehensive web-based platform designed to modernize and streamline the traditional, manual processes of the construction industry. It connects all key stakeholders—clients, consultancies, contractors, suppliers, and labours within a single unified system, enabling seamless communication and collaboration from project initiation to completion. Clients can register, search for consultancies based on their project needs, and request quotations. Consultancies, upon securing a project, can invite contractors to bid for various components. This competitive bidding process ensures better pricing and quality, while selected contractors can further assign tasks to suppliers and labours through the same interface.

The system offers features such as automated quotation management, real-time notifications, milestone-based payment tracking, and detailed project analytics. Suppliers can manage and confirm material deliveries, while labours receive assignments and report progress digitally. With role-specific dashboards, resource management tools, and transparent workflows, the platform significantly reduces delays, enhances accountability, and prevents cost overruns. By integrating every stage of the construction lifecycle into a centralized digital environment, the Construction Workflow Automation System transforms the way construction projects are planned, executed, and delivered—making the entire process smarter, faster, and more efficient.

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LIST OF ABBREVIATION

List	Abbreviations	Meaning
1	MERN	MongoDB, Express.js, React, Node.js
2	API	Application Programming Interface
3	CRUD	Create, Read, Update, Delete
4	UI	User Interface
5	CSS	Cascading Style Sheets
6	HTML	HyperText Markup Language